

UNIT -5 : SAMPLING AND SAMPLING DISTRIBUTION

Concepts:

In [statistics](#), [quality assurance](#), and [survey methodology](#), **sampling** is the selection of a subset (a [statistical sample](#)) of individuals from within a [statistical population](#) to estimate characteristics of the whole population. Two advantages of sampling are that the cost is lower and data collection is faster than measuring the entire population.

Each [observation](#) measures one or more properties (such as weight, location, colour) of observable bodies distinguished as independent objects or individuals. In [survey sampling](#), weights can be applied to the data to adjust for the sample design, particularly [stratified sampling](#).^[1] Results from [probability theory](#) and [statistical theory](#) are employed to guide the practice. In business and medical research, sampling is widely used for gathering information about a population.^[2] [Acceptance sampling](#) is used to determine if a production lot of material meets the governing [specifications](#).

The sampling process comprises several stages:

- Defining the population of concern
- Specifying a [sampling frame](#), a [set](#) of items or events possible to measure
- Specifying a [sampling method](#) for selecting items or events from the frame
- Determining the sample size
- Implementing the sampling plan
- Sampling and data collecting

Types / Methods of Sampling:

[Sampling Methods: Probability and Non-Probability Sampling](#)

Sampling methods are as follows;

- Probability Sampling
- Non-Probability Sampling

[Probability Sampling:](#)

A **probability sampling** scheme is one in which each unit in the population has a chance (greater than zero) of being selected in the sample, and this possibility can be accurately determined.

The combination of these behaviours makes it possible to produce unbiased estimations of population totals, by weighting sampled units rendering to their probability of selection.

Probability sampling may be of the following types:

Simple random sampling

In a simple random sample ('SRS') of a given size, all such subsets of the frame are given an equal probability. Each component of the frame thus has an equal probability of selection: the frame is not subdivided or partitioned. This minimizes bias and simplifies analysis of results.

Systematic sampling

Systematic sampling depend on arranging the target population according to some ordering scheme and then selecting elements at regular intervals through that ordered list. Systematic sampling involves a random start and then proceeds with the selection of every k th element from then onwards.

In this case, $k = (\text{population size} / \text{sample size})$.

It is important that the starting point is not automatically the first in the list, but is instead randomly chosen from within the first to the k th element in the list.

A simple example would be to select every 10th name from the telephone directory (an 'every 10th' sample, also referred to as 'sampling with a skip of 10').

Stratified sampling

The sampling where the population embraces several distinct categories, the frame can be organized by these categories into separate "strata." Each stratum is then sampled as an independent sub-population, out of which individual elements can be randomly selected. Dividing the population into distinct, independent strata can enable researchers to draw inferences about specific subgroups that may be lost in a more generalized random sample.

Cluster sampling

It is an example of 'two-stage sampling' or 'multistage sampling': in the first stage a sample of areas is chosen; in the second stage a sample of respondents within those areas is selected.

Multistage sampling

Multistage sampling is a complex form of cluster sampling in which two or more levels of units are embedded one in the other. The first stage consists of constructing the clusters that will be used to sample from. In the second stage, a sample of primary units is randomly selected from each cluster (rather than using all units contained in all selected clusters). In following stages, in each of

those selected clusters, additional samples of units are selected, and so on. All ultimate units (individuals, for instance) selected at the last step of this procedure are then surveyed. This technique, thus, is essentially the process of taking random samples of preceding random samples. It is not as effective as true random sampling, but it probably solves more of the problems inherent to random sampling. Moreover, It is an effective strategy because it banks on multiple randomizations. As such, it is extremely useful. Multistage sampling is used frequently when a complete list of all members of the population does not exist and is inappropriate. Moreover, by avoiding the use of all sample units in all selected clusters, multistage sampling avoids the large, and perhaps unnecessary, costs associated traditional cluster sampling.

Non-Probability Sampling:

Non-probability sampling is any sampling technique where some elements of the population have no definite chance of selection, or where the probability of selection can't be correctly determined.

Non-probability sampling may be of the following types:

Quota sampling

in quota sampling the population is first segmented into mutually exclusive sub-groups, just as in stratified sampling. Then judgment is used to select the subjects or units from each segment based on a specified proportion. It is this second step which makes the technique one of nonprobability sampling. In quota sampling the selection of the sample is non-random. For example, interviewers, might be tempted to interview those who look most helpful. The problem is that these samples may be biased because not everyone gets a chance of selection. This random element is its greatest weakness and quota versus probability has been a matter of controversy for many years.

Convenience sampling (grab or opportunity sampling):

Convenience sampling is a type of non-probability sampling which involves the sample being drawn from that part of the population which is close to hand? That is, a sample population selected because it is readily available and convenient. It may be through meeting the person or including a person in the sample when one meets them or chosen by finding them through technological means such as the internet or through phone. The researcher using such a sample cannot scientifically generalize about the total population from this sample because it would not be representative enough. For example, if the

interviewer was to conduct such a survey at a shopping centre early in the morning on a given day, the people that he/she could interview would be limited to those given there at that given time, which would not represent the views of other members of society in such an area, if the survey was to be conducted at different times of day and several times per week. This type of sampling is most useful for pilot testing.

Estimation:

It is defined as the statistical method of estimating the population parameter from the sample observation drawn from the population with the desired degree of precision. It helps to obtain a guess or estimate of the unknown true value from the sample data or past experience. The estimation of population parameters like mean, variance, proportion etc. from the corresponding sample statistics with reasonable accuracy are very common problems of statistical inference. That's why the sampling distribution of a statistic and its standard error plays a vital role in both the estimation of parameters and the testing of hypothesis. A sample statistic used to estimate a population parameter is called an estimator. Example The sample mean $\bar{X}$ is an estimator of the population mean (μ).

A specific observed numerical value of the estimator is called an estimate. For example, A random sample 200 students of Trichandra college give a mean height of 5.5ft with S.D of 0.25 ft, then if we use these specific values to estimate the mean height and S.D of the whole students,

the values 5.5ft and 0.25 ft would be estimates of population mean and population standard derivation respectively.

The estimation of the population parameters can be done in either of the following two ways.

1. Point estimation.
2. Interval estimation.

Point estimation:

The procedure of estimating an unknown parameter from a single value is known as point estimation. It provides an exact value of the unknown parameter to be estimated under investigation. A single number that is used to estimate an unknown population parameter is called points estimate.

For example.

A sample mean $\bar{X}$ and a sample standard deviation (S) are point estimates of the population mean (μ) and the population standard deviation (σ) respectively.

The number of enrollments in next year in BSC I year estimates by Trichandra college would be 900 is a point estimate. This process is called point estimation.

In this process, the main difficulty is to choose the best estimator for a parameter some criterion or proportions are needed to choose the best

estimator among the possible estimators. The desired properties are the following . They are as follows.

1. Un-biased.\
2. Consistency
3. Sufficiency
4. Efficiency.

Interval estimation

This process of estimating an unknown population from a range of values is known as interval estimating. In point estimation, we give a range value while in interval estimation we give an interval to estimate a parameter. In interval estimation, a sample to the sample variation is also taken into consideration through standard error.

A range of values used to estimate a population parameter is called an interval estimation. There are two ways for indicating an error by the extent of its range and by the probability of the true population parameter lying in that range. In this estimation, the researcher or decision maker has a better idea of the reliability of his estimate.

For example.

The number of enrollments in BSc. I year in next year estimates by Trichandra campus would be 700 to 1100 is an interval estimate. This process of estimation is called interval estimation.

Point estimation sometimes may not exactly represent the population parameter to overcome. This drawback we need, a range of values to

estimate the population parameter. As a rule of a point, estimate fluctuates from sample to sample interval estimates are preferable to point estimates. Also, the interval estimates indicate the accuracy of an estimate because the interval estimates provide a measure of the degree of uncertainty in terms of probability attached to the interval.